

mapping_id	source_ontology_id	source_entity_typ
<i>Stable mapping identifier, e.g., DPPO_CEON_0001</i>	<i>DPP/DPPO/CEON</i>	<i>Class/</i> <i>ObjectProperty/</i> <i>DataProperty/</i> <i>AnnotationPropert</i> <i>y/Individual</i>
DPPO_CEON_0002	DPPO	Class
DPPO_CEON_0003	DPPO	Class
DPPO_CEON_0004	DPPO	Class
DPPO_CEON_0005	DPPO	ObjectProperty

DPPO_CEON_0006	DPPO	ObjectProperty
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DPPO_CEON_0010	DPPO	Class
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DPPO_CEON_0011	DPPO	ObjectProperty
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DPPO_CEON_0012	DPPO	ObjectProperty
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DPPO_CEON_0013	DPPO	ObjectProperty
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DPPO_CEON_0015	DPPO	Class
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source_iri	source_prefix	source_label
<i>Absolute IRI</i>	<i>source prefix</i>	<i>Human-readable label</i>
http://w3id.org/dppo/ontology/dpp-core/Material	dpp-core	Material
http://w3id.org/dppo/ontology/dpp-core/Substance	dpp-core	Substance
http://w3id.org/dppo/ontology/dpp-comp/SubstanceOfConcern	dpp-comp	SubstanceOfConcern
http://w3id.org/dppo/ontology/dpp-odp/hasPart	dpp-odp	hasPart

http://w3id.org/dppo/ontology/dpp-odp/hasPart	dpp-odp	hasPart
http://w3id.org/dppo/ontology/dpp-info/DPPInformation	dpp-info	DPPInformation
http://w3id.org/dppo/ontology/dpp-info/containsInformation	dpp-info	containsInformation
http://w3id.org/dppo/ontology/dpp-odp/describes	dpp-odp	describes
http://w3id.org/dppo/ontology/dpp-info/isAbout	dpp-info	isAbout
http://w3id.org/dppo/ontology/dpp-odp/DPP	dpp-odp	DPP

target_ontology_i	target_entity_type	target_iri
DPP/DPP ⁺ /CEON	Class/ ObjectProperty/ DataProperty/ AnnotationProperty/ Individual	Absolute IRI
CEON	Class	http://w3id.org/CEON/ontology/material/Material
CEON	Class	http://w3id.org/CEON/ontology/material/ChemicalSubstance
CEON	Class	http://w3id.org/CEON/ontology/material/ChemicalEntity
CEON	ObjectProperty	http://w3id.org/CEON/ontology/product/hasProductComponent

CEON	ObjectProperty	http://w3id.org/CEON/ontology/product/hasProductObjectComponent
CEON	Class	http://w3id.org/CEON/ontology/resourceODP/Information
CEON	ObjectProperty	http://w3id.org/CEON/ontology/resourceODP/containsInformation
CEON	ObjectProperty	http://w3id.org/CEON/ontology/resourceODP/isAbout
CEON	ObjectProperty	http://w3id.org/CEON/ontology/resourceODP/isAbout
CEON	Class	http://w3id.org/CEON/ontology/resourceODP/Information

target_prefix	target_label	alignment_relation	direction
<i>target_prefix</i>	<i>Human-readable label</i>	<i>Controlled vocabulary (see README)</i>	<i>source_to_target/ target_to_source/ bidirectional</i>
ceon-material	Material	subclass_of	source_to_target
ceon-material	ChemicalSubstance	subclass_of	target_to_source
ceon-material	ChemicalEntity	subclass_of	source_to_target
ceon-product	hasProductComponent	subproperty_of	target_to_source

ceon-product	hasProductObjectComponent	subproperty_of	target_to_source
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resourceODP	Information	subclass_of	source_to_target
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resourceODP	containsInformation	subproperty_of	source_to_target
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resourceODP	isAbout	subproperty_of	source_to_target
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resourceODP	isAbout	subproperty_of	source_to_target
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resourceODP	Information	subclass_of	source_to_target
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mapping_notes & justification

rationale, ignored by generator for logic

DPPO defines Material as a subclass of Product, by its commercial status. CEON defines Material as a subclass of Matter, requiring at least one associated chemical entity. The two are not interchangeable, so the DPPO class is aligned as a subclass of the CEON one rather than equated with it. The DPPO material class is related to the same CEON class conceptually, which DPPO uses Substance for basic chemical entities (elements or compounds) that might appear in a product's composition or as regulated substances. CEON explicitly defines ChemicalSubstance and ChemicalElement classes. This can be seen as a specialization of DPPO's Substance class.

CEON does not have built-in classes for "substance of concern," but DPPO defines these as special Substance subclasses per EU regulation. We align this by treating DPPO's concern subclasses as subclasses of CEON's ChemicalEntity or ChemicalSubstance. For example, map `dppo:ToxicSubstance` as subclass of `ceon:material:ChemicalSubstance`, and DPPO uses one generic hasPart for products; CEON splits into model- and item-level shortcuts. Making both CEON properties subproperties of DPPO's hasPart preserves DPPO queries and CEON's extra precision. DPPO also infers partonomy from reified composition (see below).

CEON distinguishes component relations at two levels: `hasProductComponent` relates product models, while `hasProductObjectComponent` relates individual product objects. DPPO provides only the generic `hasPart`, whose values are constrained by the type of the subject rather than by granularity. The item-level CEON property is DPPO's `DPPInformation` is a piece of information contained in a DPP and about a product; CEON's `resourceODP:Information` is the general notion of DPPO relates a DPP to the pieces of information it contains; the CEON property is the general containment relation between a collection of information and its parts. Stated in the Digital Product Passport as Document section of the accompanying paper.

Expressed as a subsumption rather than an equivalence to prevent unintended inferences arising from the domain restriction of `dpp-odp:describes`. Stated in the Digital Product Passport as Document section of the accompanying paper. ~~Generalised DPPO~~ CEON_0012: `dpp-odp:describes` is declared a subproperty of `dpp-info:isAbout` within DPPO, and both `isAbout` properties relate a piece of information to the entity it describes. Additional mapping, not discussed in the paper.

CEON defines no passport class. Interpreted within the CEON framework the DPPO passport is a specialised information entity, so it is aligned as a subclass of the CEON information class. An equivalence would be incorrect, because the CEON class covers far more than passports. Stated in the section on the semantic alignment of core classes in the accompanying